

Software Engineering for Systems Modeling

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1.1 V-Model

Requirements Analysis -> High level design -> Detailed specifications -> Coding -> Unit testing
-> Integration testing -> Operational testing

Requirements can be divided into that of problem (stakeholder) and solution (system) requirements. The requirements of the problem and stakeholder are concerned with the satisfaction of the output of the system. It is concerned with what a stakeholder would like to have from the system and how they might interact with it. The requirements of the solution and system are those that lie in the bounds of the system and define what it does.

1.2 MBSE

Model-based Systems Engineering defines a standard comprehensive perspective of a system and develops a single model for the process. It lowers the cost of changes throughout the process and establishes clear boundaries of both components of the system and the processes associated.

1.3 SysML

SysML shares overlap with UML, a modeling language and standard that provides a visual standard for components of software that interact with each other. With a modeling language, we can construct concrete and visual representations of previously abstract ideas. A system could be represented as a list of documents describing everything in detail. However, a model enables visual representations more flexible and comprehensible.

SysML supports the specification, analysis, design, verification, and validation of systems that include hardware, software, data, personnel, procedures, and facilities. It is a visual modeling language that provides semantics and notations for models, which at times, can be quite tedious and has inhibited its prevalence.

The four pillars of SysML include:

1. Structure
2. Behavior
3. Requirements
4. Parametrics

A SysML drawing has a few components

- Analysis diagram
- Internal block diagram
- Package diagram
- Parametric diagram
- Requirement diagram
- Sequence diagram
- State machine diagram
- Use case diagram
- Allocation tables

Some important questions for defining a system:

- What is its purpose and how is its purpose accomplished?
- How does it interact with its environment?
- Who is the user and how do they use it?
- What are the components necessary to make it function, how do they interoperate and how do they interact with the environment?
- What are the capabilities of the system and how are they tested?
- What is the cost of the system?
- What is the lifespan of the system?

The use cases of a system define the objectives of that system.

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We tend to start a project with a use case, and define it by what it must accomplish.

Everything done in SysML follows a hierarchy.

There are **block definition diagrams** which define replicable blocks and their relationships with each other, and **internal block diagrams** which define the parts and components inside the block that define the interactions with the other blocks.

The difference between an activity and an action is that an activity is a composite of actions, whereas an action cannot be drilled down into or split into further actions.